

ZELIN TAN

tanzl@mail.ustc.edu.cn · [Homepage](#) · [Google Scholar](#)

EDUCATION

University of Science and Technology of China, Artificial Intelligence, <i>Ph.D. Student</i>	2025.6 - 2028.6
University of Science and Technology of China, Computer Technology, <i>M.Eng.</i>	2022.9 - 2025.6
University of Science and Technology of China, Computer Science <i>B.Eng.</i>	2018.9 - 2022.6

INTERNSHIP

Shanghai AI Lab: Agent&LLM Post Training Mentors: [Chen Zhang](#) and [Shuyue Hu](#) 2025.7 - Now

- **RSI4Infra – Self-Evolving AI Infrastructure**: Developing a parallel-search framework that jointly evolves the AI-infrastructure development harness and the infrastructure itself via evidence-driven meta-synthesis.
- **Enhancing LLM Skill-Use Capability**: Designed a data pipeline synthesizing long-horizon skill-dependent tasks and their multi-turn solution trajectories to improve the LLM’s Skill-use ability.
- **Reasoning-Paradigm Routing for LLM Agents**: Proposed *Select-then-Solve*, a lightweight router that picks a paradigm(ReAct, Plan-Execute etc.) before solving, improving both accuracy and inference efficiency.
- **Rubric-based RL Post-Training for LLMs**: Via decoupled normalization of the advantage function, addressed reward-signal exhaustion in conventional RLVR and reward hacking in Rubric RLVR.
- **RL Scaling Laws for LLMs**: Studied scaling laws of GRPO-based RL post-training for LLM mathematical agents across model sizes and hyperparameter settings.
- **Agent-Human discussion forum**: We present AgentPanel, a multi-agent forum for human–AI collaboration in scientific exploration.
- **Survey on LLM-based Agents**: Surveyed frontier progress of agentic reinforcement learning across games, interaction, and tool use, producing a survey report.

Meituan: LLM Inference Optimization Mentors: [Yineng Zhang](#) and [Ke Bao](#) 2024.5 - 2024.7

- **Block-granularity automatic prefix caching**: Learning to manage GPU memory via a block-based prefix tree to enable automatic prefix caching.
- Operator fusion for Generative Recommendation(GR) models based on Tensor-RT

Tencent: LLM Inference Optimization Mentor: [Hande Dong](#) 2023.12 - 2024.5

- **Chunked prefill** with dynamic split-fuse on internal serving engine (based on vLLM v0.2.7)
- **SM-Aware Operator Parallelism Framework**: Leveraged intra-GPU compute-resource isolation to run prefill and decode attention kernels in parallel.
- **GPU-memory-aware dynamic chunk-size adjustment**, mitigating the recomputation overhead introduced by pre-emption under high load.

PUBLICATIONS

-
- [SKT: Skill-Use Training at Scale via Verified Synthetic Data Generation](#) (**First author, AAAI 2027 under review**)
 - [PAPO: Stabilizing Rubric Integration Training via Decoupled Advantage Normalization](#) (**First author, EMNLP 2026 Main**)
 - [Scaling Behaviors of LLM Reinforcement Learning Post-Training: An Empirical Study in Mathematical Reasoning](#) (**First author, ACL 2026 Main**)
 - [AgentPanel: Toward a New Paradigm for Human-AI Collaboration in Exploring Scientific Questions](#) (**DAI 2026 Oral**)
 - [Scaling the Horizon, Not the Parameters: Reaching Trillion-Parameter Performance with a 35B Agent](#) (**Public Technical Report**)
 - [Parametric Skills](#) (**ICLR 2027 under review**)
 - [Select-then-Solve: Paradigm Routing as Inference-Time Optimization for LLM Agents](#)
 - [SkillsBench: Benchmarking How Well Agent Skills Work Across Diverse Tasks](#) (**NeurIPS 2026**)
 - [AgentAsk: Multi-Agent Systems Need to Ask](#) (**ACL 2026 Main**)
 - [The Landscape of Agentic Reinforcement Learning for LLMs: A Survey](#) (**TMLR 2025**)

OPEN SOURCE

[ROLL](#) (contributor) · [AgentPanel](#) (contributor) · [SkillsBench](#) (contributor) · [libsmctrl](#) (designer)